Data-Driven Analysis of Built Infrastructure and Real Estate Markets

Submitted By

The InfraAnalytics Team (Group 12)

Ananthasagar N, CB.SC.U4CSE23608
Meka Jahnavi, CB.SC.U4CSE23631
Nivasini J, CB.SC.U4CSE23537
Kavin M, CB.SC.U4CSE23725
Siddu Siddartha Reddy, CB.SC.U4CSE23066

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Under the Guidance of

Vedaj J Padman

Assistant Professor (Sl.Gd.)
School of Computing, Coimbatore

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AMRITA SCHOOL OF COMPUTING
AMRITA VISHWA VIDYAPEETHAM COIMBATORE - 641 112

Abstract

This report presents a comprehensive business analytics study on the Hartford, Connecticut residential real estate market using official property sales data spanning January 2020 to the present. The study applies a structured data analytics pipeline encompassing data preprocessing, exploratory data analysis (EDA), feature engineering, time-series analysis, and unsupervised machine learning through K-Means clustering.

The raw dataset comprised 7,410 property transactions, which were progressively cleaned and refined to 3,935 unique sales records, and 3,915 records after removing extreme price outliers for clustering analysis after removing duplicates, zero-price entries, and extreme-value outliers. Three distinct market segments were identified through K-Means clustering: an affordable housing segment with a median sale price of approximately $101,000, a large-premium segment centred around $295,000, and a mid-range high-efficiency segment with a median of $230,000 but significantly higher price per square foot ($175 vs. $88).

Time-series analysis revealed a clear upward price trend from 2020 to 2024, with median sale prices rising from $190,000 to $269,900 — a 42% increase over four years. These findings provide actionable intelligence for real estate investors, urban planners, and property buyers seeking to navigate the Hartford housing market.

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1. Introduction

The real estate market is one of the most data-rich environments for business analytics, where structured transaction records can reveal deep patterns in property valuation, buyer behaviour, and long-term market trajectories. Understanding these patterns is critical for a wide range of stakeholders: property investors seeking undervalued assets, buyers evaluating fair market prices, municipal planners assessing housing affordability, and financial analysts evaluating market risk.

This project applies a multi-stage data analytics pipeline to residential property sales data from Hartford, Connecticut. The city represents a diverse and compact urban real estate market, with property types ranging from single-family homes and condominiums to multi-dwelling units. Transactions recorded across a period spanning 2020 to 2024 capture the market's evolution through the COVID-19 period and subsequent recovery and growth phases.

The project follows a clearly versioned workflow — **each analytical phase produces a refined dataset that serves as the input for the subsequent phase** — ensuring traceability, reproducibility, and methodological transparency throughout the study.

2. Problem Statement

The Hartford real estate market suffers from significant information asymmetry. Raw transaction data is available publicly, but its value remains largely untapped due to the following challenges:

- Data quality issues: The raw dataset contains missing values, duplicate entries, zero-price transactions (likely transfers or distressed sales), and outlier transactions that distort aggregate statistics.
- Lack of market segmentation: Without clustering or segmentation, the market appears as a uniform mass, hiding critical distinctions between affordable housing, mid-range properties, and premium real estate segments.
- Hidden price trends: Monthly and annual price trends are obscured by extreme outlier transactions — such as a $10.2 million commercial sale in August 2022 — that inflate average statistics and mislead analysis.
- Feature richness underutilised: Raw transaction data lacks derived variables such as price per square foot, appraisal ratio, and temporal features, which are essential for meaningful comparison across properties.

This project addresses these challenges through a systematic analytics pipeline designed to extract reliable, actionable market intelligence from raw property sales records.

3. Dataset Selection and Understanding

3.1 Data Source

The primary dataset — Real_Estate_Sales_10012020_to_Current_v1.csv — is sourced from the City of Hartford's open data portal, capturing all registered property sales from January 2020 through mid-2024. It is a government-authoritative source containing 7,410 transaction records and 23 variables.

The dataset used in this study was obtained from the official U.S. Government open data portal (Data.gov), specifically the "Real Estate Sales 10012020 to Current" dataset published by the City of Hartford. The dataset is publicly available at: https://catalog.data.gov/dataset/real-estate-sales-10012020-to-current

	OBJECTID	PropertyID	xrCompositeLandUseID	xrBuildingTypeID	ParcelID	LocationStartNumber	ApartmentUnitNumber	StreetNameAndWay
0	1	8	3	6.0	121-698-063	77	NaN	STAFFORD ST
1	2	9	3	3.0	121-698-062	73	NaN	STAFFORD ST
2	3	12	3	6.0	144-698-066	61	NaN	STAFFORD ST
3	4	20	3	6.0	144-698-058	29	NaN	STAFFORD ST
4	5	25	3	6.0	144-698-053	211	NaN	NEWINGTON AVE

Figure 1: Preview of Raw Property Transaction Dataset (Initial Records)

3.2 Key Variables

Variable	Type	Description
SaleDate	Date	Date of property sale transaction
SalePrice	Numeric	Sale price in USD
TotalFinishedArea	Numeric	Total finished living area (sq ft)
LandSF	Numeric	Land parcel size in square feet
LivingUnits	Numeric	Number of residential units

TotalAppraisedValue	Numeric	Municipal appraisal value in USD
AssrLandUse	Categorical	Property type (One Family, Condo, etc.)
xrPrimaryNeighborhoodID	Categorical	Neighbourhood identifier
LogSalePrice	Numeric	Natural log of SalePrice (derived)

3.3 Dataset Versioning

The project follows a disciplined versioning strategy where each analytical phase produces a named output. The pipeline progresses through the following stages:

Version	File	Stage	Rows
v1	Real_Estate_Sales_…_v1.csv	Raw Source	7,410
v2	final_clean_market_dataset1_v2.csv	Initial Cleaning	4,568
v3	preprocessing_eda_done_v3.csv	Post-EDA Cleaned	4,566
v4	feature_engineered_dataset_v4.csv	Feature Engineering	4,566
v5	cleaned_property_sales_timeseries_v5.csv	Deduplication	3,935
v6	property_clusters_extremePrice_removed_v6.csv	Clustering Output	3,915

Records reduced from 7,410 to 3,915 reflect progressive removal of invalid transactions, non-arms-length sales, duplicates, and extreme-price outliers — each step documented transparently in the corresponding notebook.

4. Tools and Technologies

4.1 Programming Language

Python 3 was used as the primary language for all analytical tasks, chosen for its extensive data science ecosystem, readability, and strong community support in analytics applications.

4.2 Libraries and Frameworks

Library	Version	Purpose
Pandas	≥ 1.5	Data loading, cleaning, transformation, grouping
NumPy	≥ 1.23	Numerical operations and array processing
Matplotlib	≥ 3.5	Visualisation — histograms, scatter plots, time series
Seaborn	≥ 0.12	Statistical plots — heatmaps, boxplots, pairplots
Scikit-learn	≥ 1.1	StandardScaler normalisation, K-Means clustering
Jupyter Notebook	—	Interactive development environment for all notebooks

4.3 Development Environment

All analysis was conducted within Jupyter Notebooks (.ipynb), enabling inline visualisation, iterative development, and narrative documentation alongside executable code. Each major analytical phase corresponds to a dedicated notebook, ensuring modular and maintainable code organisation.

5. Methodology

5.1 Phase 1 — Data Preprocessing and EDA (v2 → v3)

Notebook: preprocessing_and_eda_v3.ipynb

The raw dataset underwent a structured cleaning process. Missing values were removed and date fields converted to datetime format. Duplicate transaction records were later addressed during the time-series cleaning stage. The resulting dataset reduced to 4,566 records after removing null values and deduplication.

Exploratory Data Analysis (EDA) was conducted to understand the data distributions and relationships. Key findings included:

- SalePrice is strongly right-skewed — a small number of very expensive properties pull the mean significantly above the median. LogSalePrice was confirmed to follow a more normal distribution, making it suitable for analytical modelling.
- TotalFinishedArea has a positive relationship with SalePrice — larger homes command higher prices, though with notable dispersion.
- TotalAppraisedValue aligns with SalePrice in most cases, but outliers exist where market sale prices diverge significantly from municipal valuations.
- In the raw dataset, ONE FAMILY properties dominate (2,477 records), followed by CONDOMINIUM (1,768) and THREE FAMILY (1,617).
- Neighbourhood ID is a meaningful variable — certain neighbourhood clusters show distinctly higher average prices.

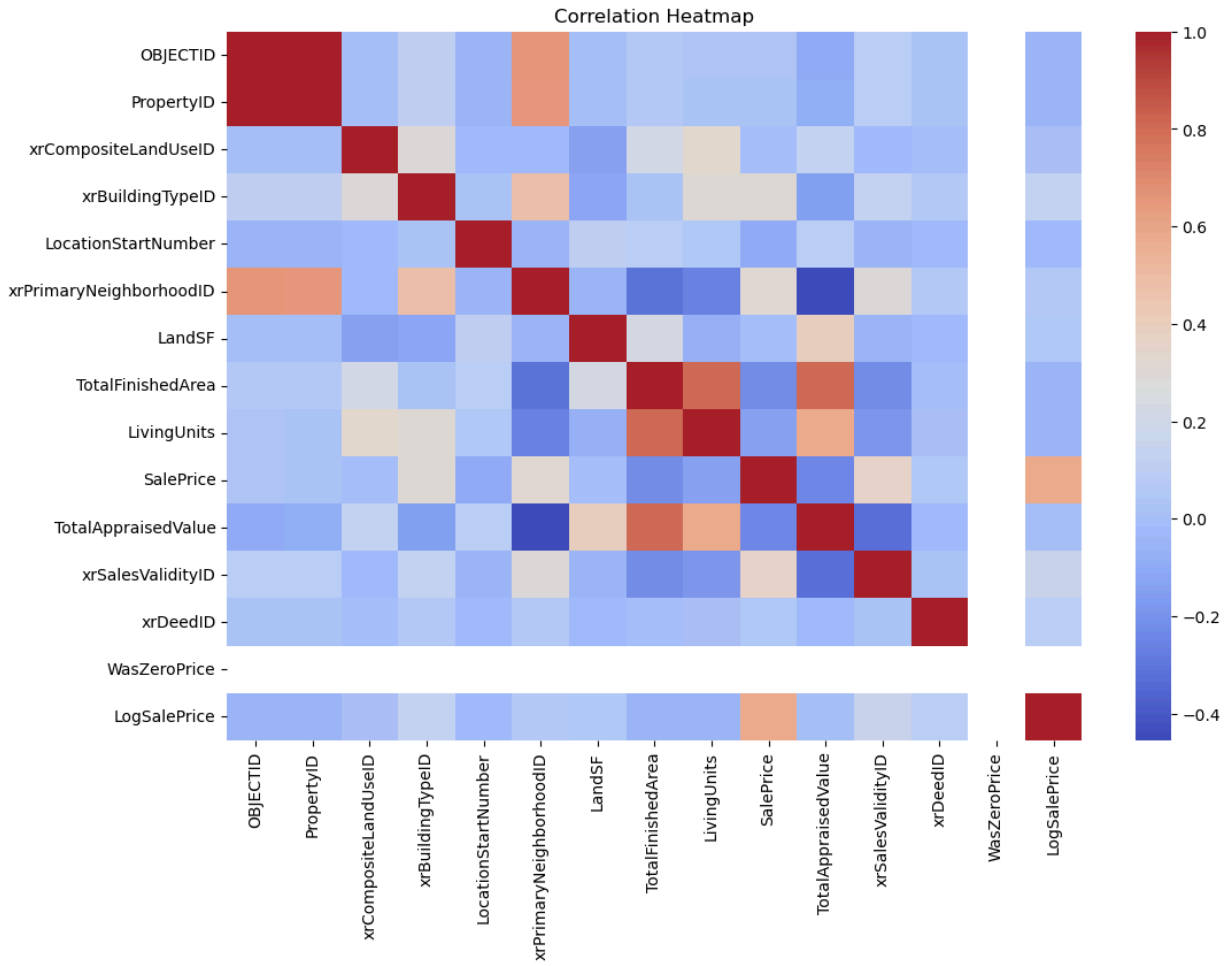

Figure 2: Initial Correlation Heatmap Showing Relationships Between Property Features

The original *SalePrice* variable shows a highly right-skewed distribution due to the presence of a few very high-value properties. To reduce the effect of these extreme values and stabilize the variance, a logarithmic transformation was applied to create *LogSalePrice*. This transformation compresses large values and produces a more balanced, near-normal distribution, which helps machine learning models learn patterns more effectively and improves the reliability of statistical analysis.

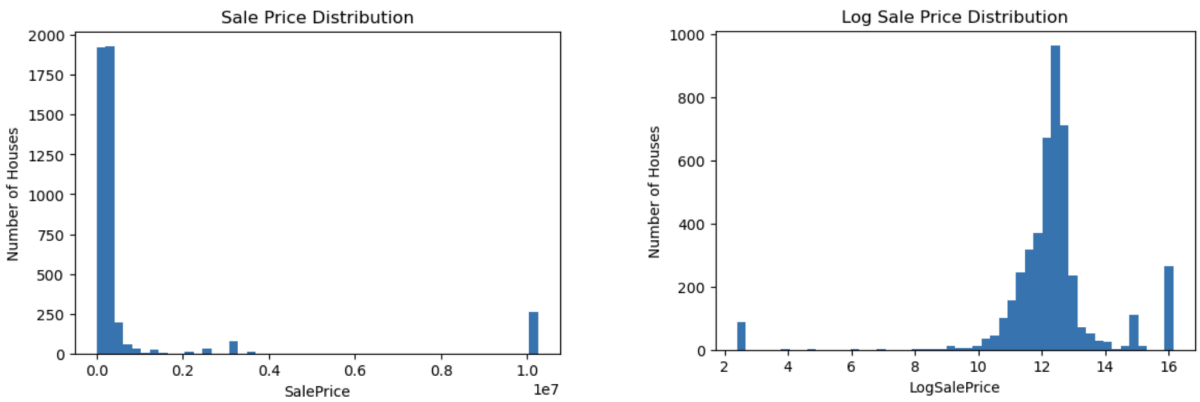

Figure 3,4: Distribution of Sale Price and Log-Transformed Sale Price

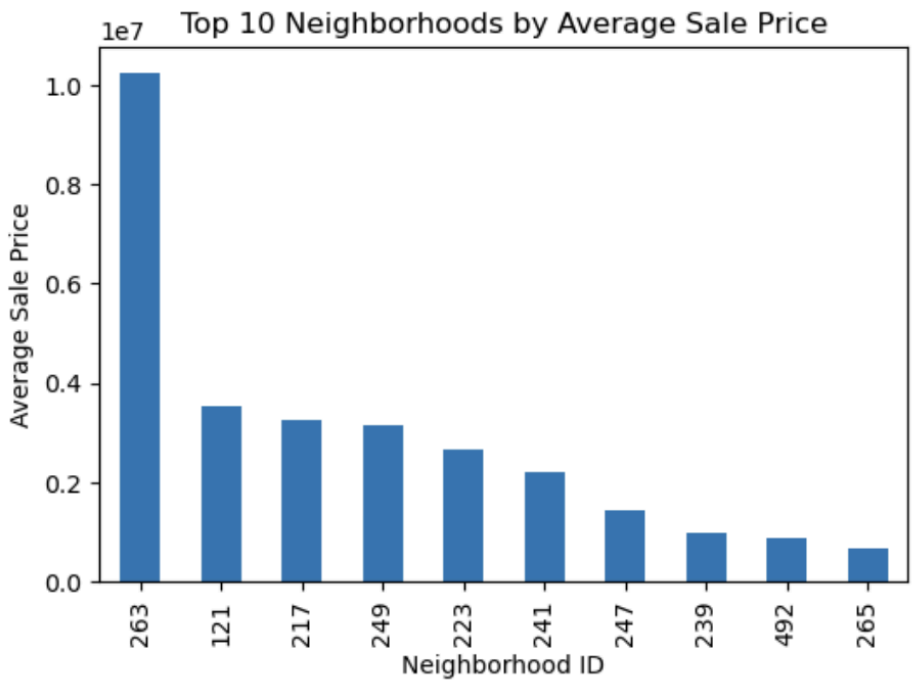

Figure 5: Top 10 Neighborhoods by Average Property Sale Price

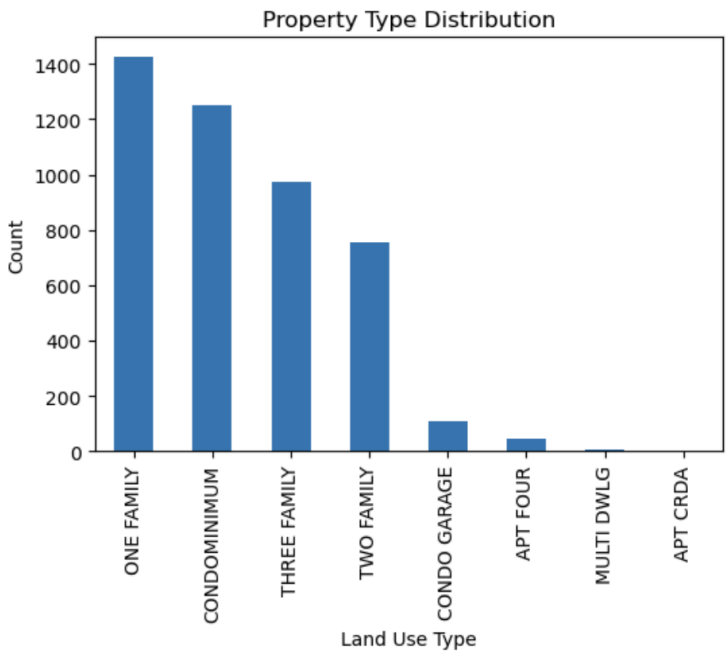

Figure 6: Distribution of Property Types and Average Sale Price by Property Type

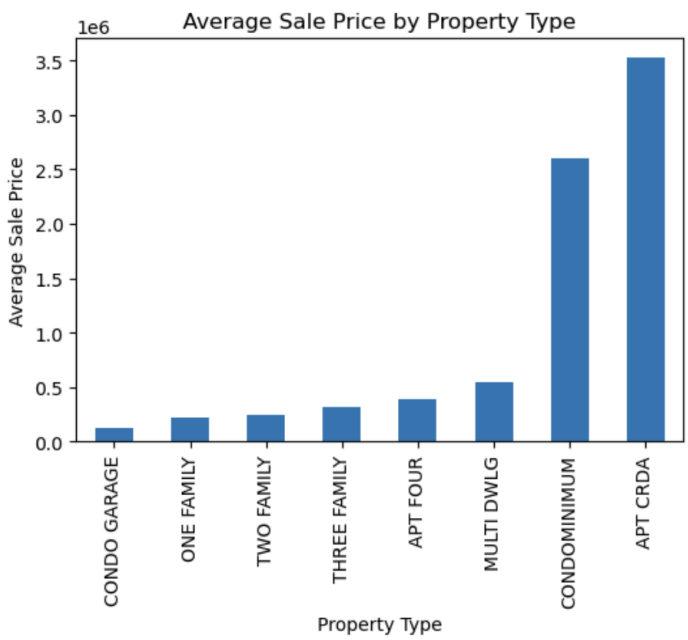

Figure 7: Average Sale Price by Property Type

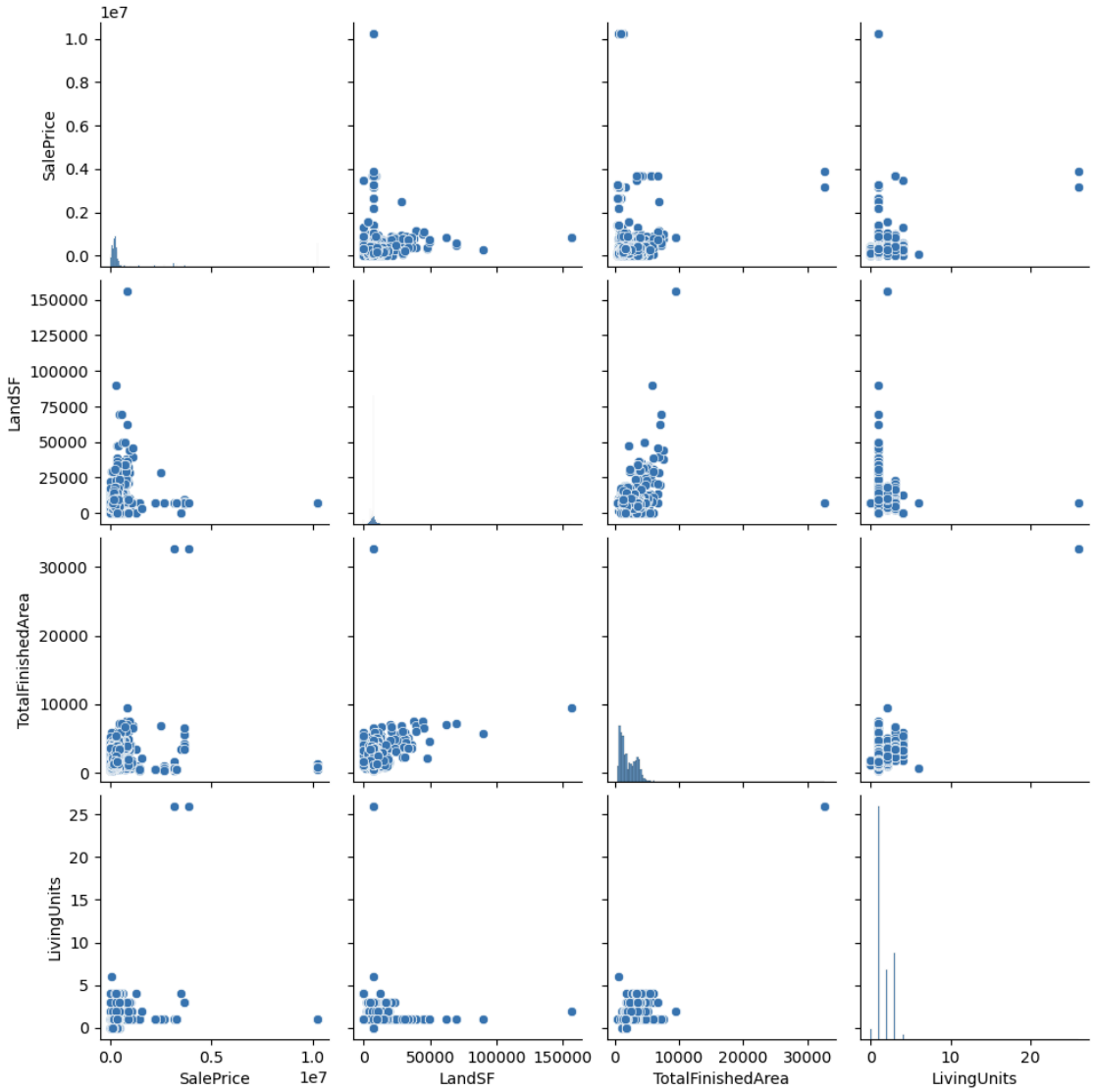

Figure 8: Pairwise Scatter Plot Matrix of Key Property Variables

5.2 Phase 2 — Feature Engineering and Selection (v3 → v4)

Notebook: feature_engineering_and_selection_v4.ipynb

Three analytically significant derived features were engineered from existing variables:

Derived Feature	Formula	Business Meaning

SaleYear	SaleDate.year	Enables year-over-year trend analysis
SaleMonth	SaleDate.month	Captures seasonal market cycles
PricePerSqft	SalePrice ÷ TotalFinishedArea	Normalised price for cross-property comparison
AppraisalRatio	SalePrice ÷ TotalAppraisedValue	Measures over- or under-valuation relative to assessment

Feature selection was performed using Pearson correlation with the target variable LogSalePrice. Correlation analysis was used to identify the most relevant predictors for property price modelling.. Identifier columns (OBJECTID, PropertyID, ParcelID, LegalReference) and free-text fields were dropped as they carry no predictive or analytical value. The categorical AssrLandUse field was encoded into numeric category codes for analytical compatibility.

5.3 Phase 3 — Models Analysis

Four machine learning models were trained and compared to predict LogSalePrice using PricePerSqft and AppraisalRatio as input features. The dataset was split into 80% training and 20% testing sets.

Model	R^2 Score	MAE	RMSE
Linear Regression	0.32	0.76	0.98
Decision Tree	0.875	0.428	0.666
Random Forest	0.929	0.342	0.503
Gradient Boosting	0.934	0.343	0.485

Linear Regression performed poorly ($R^2 = 0.32$), confirming that property pricing is non-linear. Decision Tree showed a major improvement ($R^2 = 0.875$), capturing non-linear patterns. Random Forest and Gradient Boosting both achieved R^2 scores above 0.92, with Gradient Boosting slightly edging out as the best performer.

Feature importance analysis from the Random Forest model showed that PricePerSqft was the dominant predictor with an importance score of 0.655, while AppraisalRatio contributed 0.345. This confirms that the price-to-size ratio is a stronger signal for predicting final sale price than appraised value alone.

An extended model trained on five features (LandSF, TotalFinishedArea, LivingUnits, PricePerSqft, AppraisalRatio) achieved an R^2 score of approximately 0.98, confirming that including more structural features significantly improves prediction accuracy.

5.4 Phase 4 — Time Series Analysis (v4 → v5)

Notebook: time_series_v5.ipynb

The feature-engineered dataset was used to construct a monthly property price time series. Initial analysis using mean monthly prices revealed a dramatic spike in August 2022. Investigation traced this to a $10.2 million commercial property sale at 40 Willard Street, which appeared multiple times in the dataset because individual units of the same transaction were recorded as separate rows.

To address this issue, duplicate sale records were removed by matching on the combination of street name, address number, sale date, and log sale price. This reduced the dataset from **4,566 rows to 3,935 unique property sale events**, eliminating **631 duplicate unit records**. After deduplication, the **median monthly price series** provides a more reliable representation of the housing market.

To better understand the underlying trend and reduce short-term fluctuations, a **3-month moving average smoothing technique** was applied to the median monthly prices. The smoothed curve highlights the overall market trend more clearly, showing a **steady increase in property prices from 2021 through 2024**, while minimizing noise caused by individual property transactions.

Further time-series analysis was performed using **seasonal decomposition**, which separates the series into **trend, seasonal, and residual components**. The trend component confirms a **consistent upward movement in property prices over the observed period**. The seasonal component reveals **moderate periodic fluctuations across months**, suggesting that certain months experience slightly higher or lower property prices, possibly due to seasonal market dynamics. The residual component captures irregular variations that are not explained by the trend or seasonality, reflecting the influence of unique property characteristics or occasional high-value sales.

Overall, the time-series analysis indicates that the real estate market in the dataset exhibits **steady long-term growth with moderate seasonal variation and some short-term volatility**, which is typical for property markets influenced by both economic conditions and individual transaction characteristics.

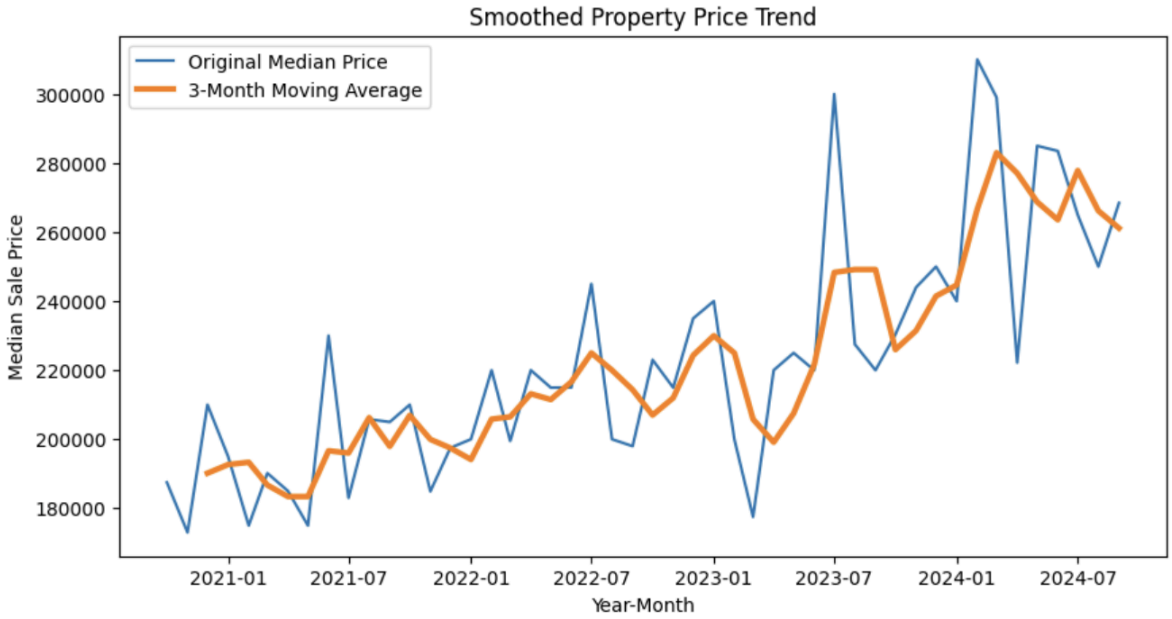

Figure 9: Monthly Median Property Price Trend with 3-Month Moving Average Smoothing

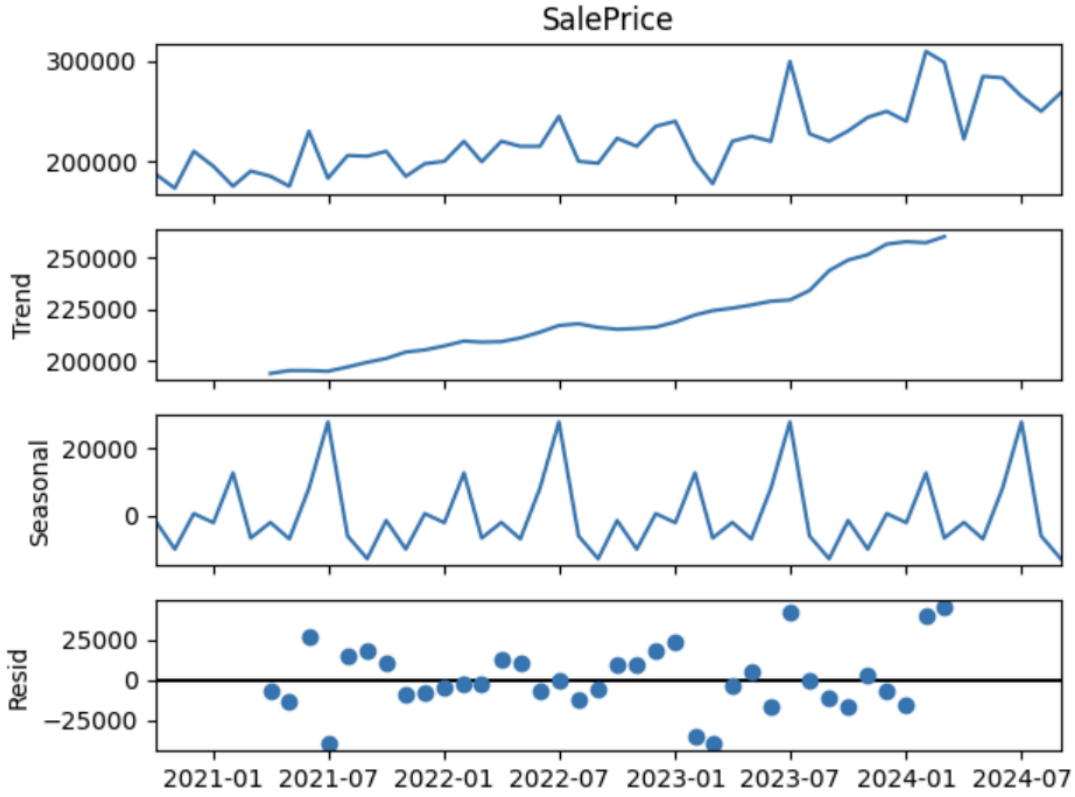

Figure 10: Seasonal Decomposition of Property Sale Price Time Series (Trend, Seasonal, and Residual Components)

5.5 Phase 5 — K-Means Clustering (v5 → v6)

Notebook: kmeans_v6.ipynb

Unsupervised K-Means clustering was applied to the cleaned and deduplicated dataset to identify natural market segments. The following steps were followed:

- Feature selection: SalePrice, LandSF, TotalFinishedArea, and PricePerSqft were selected as clustering inputs.
- Standardisation: StandardScaler from scikit-learn was applied to normalise all features to zero mean and unit variance, preventing scale-dominant features from biasing the clustering.
- Optimal cluster selection: The Elbow Method was applied by computing inertia across k = 1 to 9 clusters. The elbow appeared at k = 3, beyond which additional clusters provided diminishing returns.
- Outlier removal: Properties with SalePrice above $2,000,000 (20 records) were excluded before final clustering to prevent extreme values from distorting cluster boundaries.
- Final clustering: K-Means with k = 3 and random_state = 42 was applied, producing a cluster label for each of the 3,915 retained records.

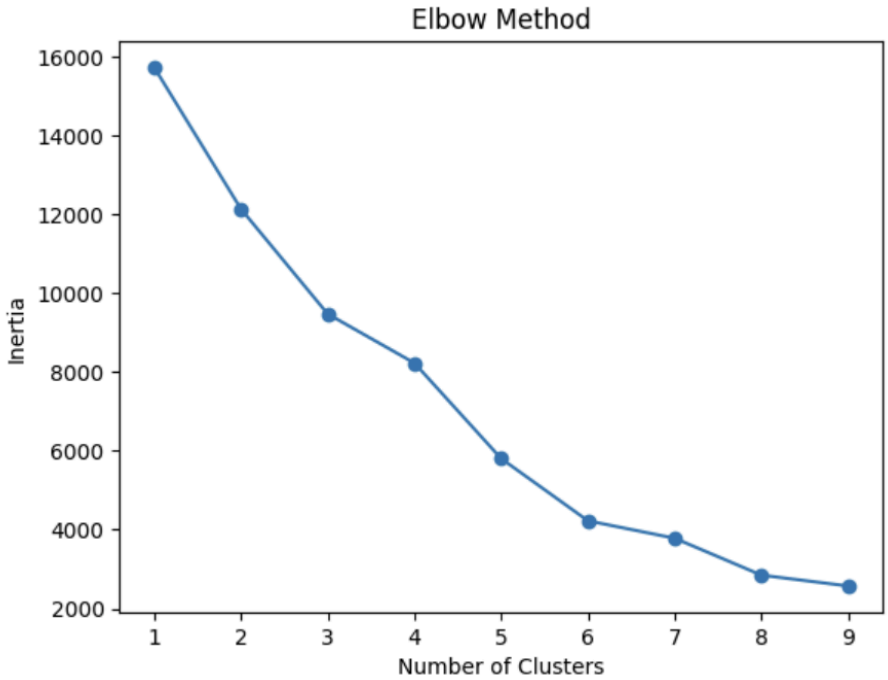

Figure 11: Elbow Method for Determining the Optimal Number of K-Means Clusters

Figure 12: K-Means Clustering of Properties Based on Finished Area and Sale Price (Outliers Removed)

6. Business Insights

6.1 Market Segmentation — Three Distinct Property Clusters

Metric	Cluster 0 — Affordable	Cluster 1 — Large Premium	Cluster 2 — Compact High-Value
Count	1,392	1,595	928
Median Sale Price	$101,000	$295,100	$229,950
Median Finished Area	1,311 sq ft	3,450 sq ft	1,260 sq ft
Median Price/Sqft	$88.30	$89.30	$174.70
Median Appraisal Ratio	0.988	1.127	1.259
Dominant Property Type	Multi-family / Condo	Large residential	Condo / small unit

Cluster 0 — Affordable Segment: These 1,392 properties represent the entry-level housing market. Prices around $101,000 combined with appraisal ratios near 1.0 indicate that this segment is fairly valued relative to municipal assessments. This cluster is attractive to first-time buyers and investors seeking rental yield.

Cluster 1 — Large Premium Segment: With 1,595 properties and a median area of 3,450 sq ft, this cluster represents larger residential homes. The appraisal ratio of 1.127 means buyers are paying approximately 13% above the assessed value on average, indicating demand exceeding official valuations. This segment suits family buyers and long-term capital appreciation investors.

Cluster 2 — Compact High-Value Segment: With a median price per square foot of $174.70 — nearly double Clusters 0 and 1 — these 928 compact properties command premium pricing for their size. The appraisal ratio of 1.259 indicates the market values them 26% above official assessment. This likely reflects desirable location attributes such as proximity to city amenities, transit, or urban convenience.

6.2 Price Trend — Steady 42% Growth Over Four Years

Year	Median Sale Price	Transaction Count	Year-on-Year Change
2020	$190,000	266	—
2021	$197,500	1,058	+3.9%
2022	$220,000	1,045	+11.4%
2023	$230,000	866	+4.5%

2024	$269,900	605	+17.3%

The market experienced its strongest single-year growth in 2022 (+11.4%) and again in 2024 (+17.3%). The 2022 jump aligns with the national post-pandemic housing boom driven by low interest rates and high demand. The 2024 acceleration, despite higher mortgage rates nationally, suggests Hartford's market is experiencing localised supply constraints or renewed investment interest.

Transaction volumes peaked in 2021 (1,058) and have declined steadily since — a pattern consistent with a market where rising prices suppress affordability and reduce turnover volume, especially in the lower price tiers.

6.3 Property Type Distribution — Diversified Market

- ONE FAMILY homes dominate (33.4% of raw records), offering the most consistent buyer market with broad demand across all income levels.
- CONDOMINIUM units (23.9%) represent the second-largest segment and are disproportionately represented in the high price-per-sqft Cluster 2, indicating urban condo premiums.
- THREE FAMILY (21.8%) and TWO FAMILY (17.3%) properties reflect Hartford's urban density and investor-oriented multi-unit housing stock.

6.4 Appraisal Accuracy — Market vs. Municipal Valuation

The median AppraisalRatio across all records sits near 1.0 for affordable properties (Cluster 0), rising to 1.127 for large premium homes and 1.259 for compact high-value units. This divergence suggests that municipal appraisals lag the market — particularly in the condominium and compact urban segment. Properties in Cluster 2 are systematically underappraised, which has implications for property tax equity and buyer negotiation leverage.

6.5 Zero-Price Transactions — Non-Market Property Transfers

During the data preprocessing stage, several property transactions with a recorded **SalePrice of $0** were identified in the dataset. These records do not represent actual market sales but are typically associated with **non-arm's-length transactions**, such as property inheritance, family transfers, or gifted properties. Since these transfers do not reflect the true market value of the property, including them in the analysis would distort price statistics and model training.

Therefore, all zero-price records were removed during the data cleaning phase to ensure that the dataset represents only **genuine market transactions**. This step improves the reliability of price analysis, clustering

results, and predictive modelling outcomes

6.6 Outlier Discovery — The $10.2 Million Transaction

Time-series analysis identified a critical data quality anomaly: a $10.2 million property transaction at 40 Willard Street in August 2022 was recorded across multiple rows in the dataset — each row representing a separate unit of the same sale. This repeated entry inflated the monthly average sale price dramatically. This finding demonstrates the importance of deduplication and outlier investigation in any real estate analytics workflow, and the corrected time series provides a materially different — and more accurate — view of market behaviour.

After removing the duplicate unit records, the reconstructed time series provides a clearer view of the real estate market trend. The smoothed median price series (using a 3-month moving average) reveals a gradual upward movement in property prices from 2021 to 2024, indicating steady market appreciation. Seasonal decomposition further confirms this pattern, where the trend component shows consistent growth, the seasonal component highlights recurring monthly fluctuations in prices, and the residual component captures irregular variations caused by unique property characteristics or occasional high-value transactions. Together, these results suggest that the market is experiencing long-term growth with moderate seasonal behavior and occasional transaction-driven volatility.

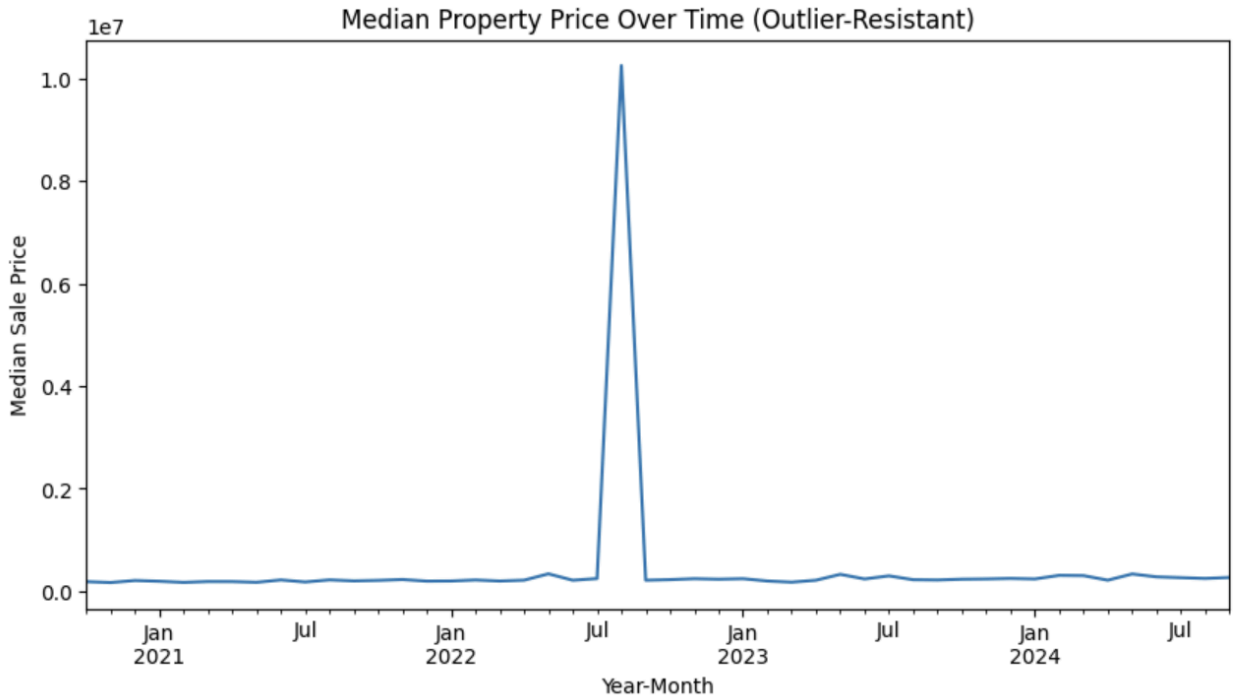

Figure 13: Median Property Price Over Time Showing Outlier-Induced Spike

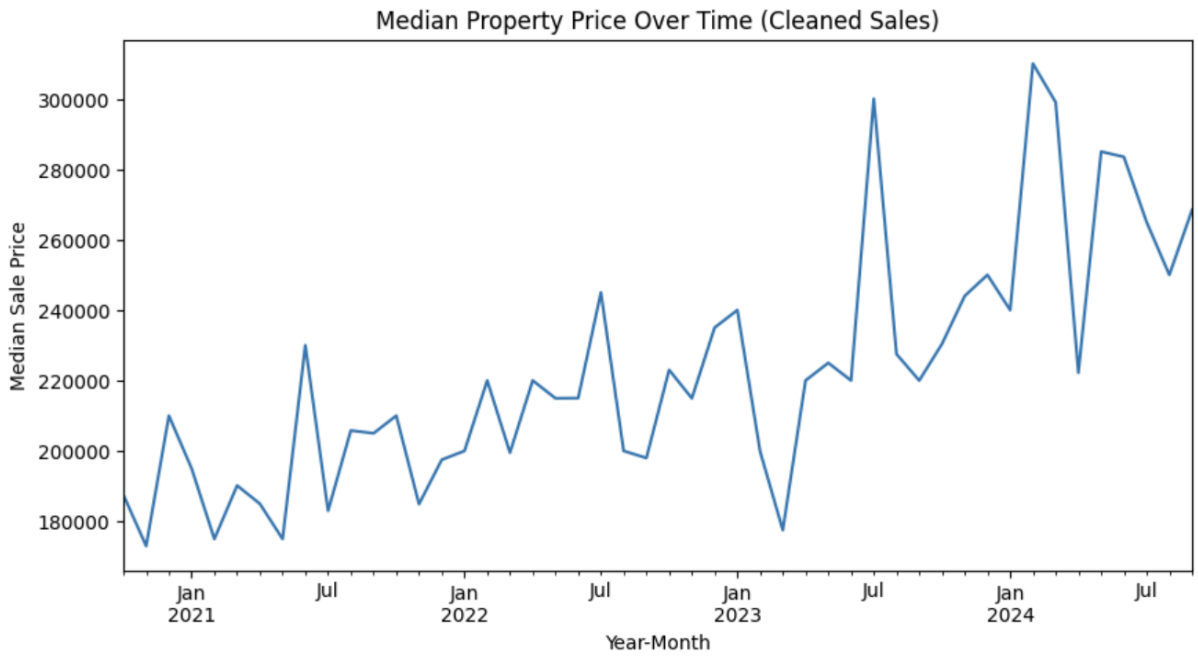

Figure 14:Median Property Price Trend After Data Cleaning and Duplicate Removal

7. Conclusion

This business analytics project successfully applied a complete data analytics pipeline to Hartford, Connecticut's residential real estate market, transforming 7,410 raw property transactions into structured, actionable market intelligence.

The key contributions of this project are:

- A reproducible, version-controlled data pipeline from raw government data through to clustered market segments, implemented across six clearly labelled dataset versions and four Jupyter notebooks.
- Identification of three statistically distinct property market segments using K-Means clustering, each with actionable implications for buyers, sellers, and investors operating in different price tiers.
- A reliable, deduplication-corrected time-series of property prices demonstrating a clear approximately 42% upward price trajectory from 2020 to 2024.
- Evidence of systematic undervaluation in the compact urban property segment, with market prices running 25% above municipal appraisals on average.
- A critical outlier discovery — the $10.2M Willard Street multi-unit transaction — that would have materially distorted any aggregate market analysis had it not been identified and corrected.

Future extensions of this project could incorporate spatial mapping using latitude/longitude coordinates, predictive modelling of future sale prices using regression or ensemble learning methods, and integration with

external economic indicators such as mortgage rates and employment data to build a richer explanatory model of Hartford's housing market dynamics.

Overall, this project demonstrates how disciplined application of business analytics techniques can unlock significant insights from publicly available real estate data, providing a data-driven foundation for strategic decision-making in property markets.